

PAPUA NEW GUINEA UNIVERSITY OF TECHNOLOGY

Department of Mathematics and Computer Science

Project Title:

Fuzzy Logic Controller for Credit Risk & Loan Approval in MSMEs/Microfinance

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Abstract

This study designs and simulates a Mamdani fuzzy inference system (FIS) to assess MSME loan applications in Papua New Guinea. Traditional credit assessment methods struggle to handle uncertainty and incomplete borrower information. Our system evaluates multiple borrower attributes to provide Loan Approval Decision (Reject, Partial Approval, Full Approval) and Credit Risk Level (High, Medium, Low). Inputs include Income Stability, Collateral Value, Business Age, Repayment History, and Loan Amount Requested. The system employs trapezoidal, triangular, and Gaussian membership functions for fuzzification, Mamdani min–max aggregation for inference, and centroid defuzzification for crisp outputs. Regression and membership function analyses indicate strong alignment between input attributes and output decisions. Sample scenarios show that the system accurately classifies risk and loan approval categories, improving interpretability and robustness compared to traditional approaches.

Keywords

Fuzzy logic; Mamdani FIS; MSME loans; consumer Credit risk; Loan approval; Membership function; PNG

1. Introduction

1.1 Problem Statement

Micro, Small, and Medium Enterprises (MSMEs) in Papua New Guinea often face challenges accessing finance due to limited credit information, fluctuating income, and subjective risk assessments. Traditional credit scoring methods fail to adequately account for uncertainty and partial information, leading to either over-cautious or risky lending decisions.

1.2 Research Gap

Existing credit assessment tools are primarily statistical and rule-based, lacking flexibility in handling fuzzy or imprecise borrower attributes. Fuzzy logic can emulate human reasoning and handle these uncertainties efficiently, yet its application to MSME loan assessment in PNG remains underexplored.

1.3 Objectives and Contributions

- Design a Mamdani FIS for MSME loan evaluation.
- Model uncertainty in borrower attributes using membership functions.
- Provide interpretable outputs: Loan Approval Decision (LAD) and Credit Risk Level (CRL).
- Conduct regression and membership function analysis to validate system accuracy.

1.4 Paper Roadmap

Section 2 reviews related work. Section 3 presents the methodology including input/output design, membership functions, rule base, and inference process. Section 4 discusses experiments and sample scenarios. Section 5 analyzes results and system performance. Section 6 concludes and suggests future research directions.

2. Related Work

Fuzzy logic has been widely applied in credit risk assessment and financial decision-making. Type-1 and Mamdani FIS have been used for SME loan approval, demonstrating superior interpretability over statistical models. Interval Type-2 fuzzy sets

have further improved robustness to input uncertainty. In PNG, limited research exists on fuzzy-based MSME credit evaluation, highlighting the need for localized systems.

3. Methodology

The methodology for this project involves designing and simulating a **Mamdani fuzzy inference system (FIS)** to evaluate credit risk and loan approval decisions for MSMEs in PNG..Figure1 show the general structure of Fuzzy System. The system uses expert knowledge encoded as **fuzzy rules** to model uncertainty in borrower attributes and generate interpretable outputs. The methodology is divided into four main sub-sections: Inputs and Outputs, Membership Function Tables, Rule Base, and Fuzzy Inference & Defuzzification.

3.1Inputs and Outputs

Inputs

Input	Description	Categories	Membership Function Type	Notes
Income Stability (IS)	Consistency of business cash-flow	Low, Medium, High	Trapezoidal	Captures cash-flow regularity and stability; higher IS implies better repayment ability.
Collateral Value (CV)	Value of assets/savings relative to loan	Poor, Moderate, Strong	Triangular	Represents security available to back the loan
Business Age (BA)	Years in operation	New, Growing, Established	Gaussian	Reflects business

				maturity; older businesses are generally more stable.
Repayment History (RH)	Past loan repayment performance	Poor, Fair, Good	Triangular	Historical repayment behavior informs future risk.
Loan Amount Requested (LA)	Size of loan requested	Small, Medium, Large	Trapezoidal	Higher loan amounts increase exposure and potential risk

Outputs

Output	Categories	Membership Function Type	Notes
Credit Risk Level (CRL)	High Risk, Medium Risk, Low Risk	Triangular	Indicates the likelihood of borrower default; used for risk assessment.
Loan Approval Decision (LAD)	Reject ,Partial Approve , Fully Approve	Trapezoidal	Provides a crisp lending decision based on input attributes and fuzzy rules.

Notes:

- The input and output selection is based on industry standards for MSME loan evaluation and prior research.
- Linguistic variables allow modeling of subjective concepts (e.g., “Low Income Stability”) that cannot be precisely quantified

3.2 Membership Function Tables

Membership functions (MFs) are a fundamental component of the fuzzy inference system, as they translate numerical input values into linguistic variables that can be processed by fuzzy rules. Each input attribute (Income Stability, Repayment History, Collateral Value, Business Age, and Loan Amount) is represented using suitable membership function types such as trapezoidal, triangular, or Gaussian, chosen for their balance between interpretability and accuracy. The outputs (Credit Risk Level and Loan Approval Decision) are similarly modeled to capture the uncertainty and gradation of real-world lending decisions.

The following tables (Table 1–5) present the definitions of input membership functions, while Tables 6 and 7 describe the output functions. The parameters (a, b, c, d) specify the support and core of each function, ensuring smooth transitions between categories and realistic decision-making behavior.

Table 1: Income Stability (Trapezoidal)

Label	a	b	c	d
Low	0	0	200	400
Medium	300	500	700	900
High	800	1000	∞	∞

Interpretation:

- *Low Income Stability* represents businesses with very irregular or insufficient cash flow, common among newly established MSMEs.
- *Medium* captures businesses with moderately consistent income, often showing growth but still exposed to volatility.
- *High* indicates strong, steady, and predictable income, typical of mature businesses with established markets

Table 2: Repayment History (Triangular)

Label	a	b	c
Poor	0	0	3
Fair	2	5	8
Good	7	10	10

Interpretation:

- *Poor* repayment history means the applicant has defaulted or delayed payments multiple times.
- *Fair* indicates a mixed record with occasional late payments but not chronic default.
- *Good* refers to borrowers with consistent repayment behavior, considered reliable by financial institutions

Table3: Collateral Value (Trapezoidal)

Label	a	b	c	d
Low	0	0	20	40
Medium	30	50	70	90
High	80	100	∞	∞

Interpretation:

- *Low* collateral means the borrower has very few assets to secure the loan, raising risk for lenders.
- *Medium* represents a fair level of asset backing, giving moderate security.
- *High* collateral reflects strong financial backing, significantly lowering the lender's exposure to loss.

Table4: Business Age (Triangular)

Label	a	b	c
New	0	0	2
Growing	1	3	6
Established	5	10	10

Interpretation:

- *New* businesses (0–2 years) often lack financial records and face high uncertainty.
- *Growing* businesses (3–6 years) are past the start-up phase and beginning to stabilize revenues.
- *Established* businesses (5+ years) have proven sustainability, making them less risky to finance.

Table5: Loan Amount (Trapezoidal)

Label	a	b	c	d
Small	0	0	50	100
Medium	80	150	250	300
Large	250	400	∞	∞

Interpretation:

- *Small* loans are typical of microfinance institutions and carry low lender risk.
- *Medium* loans reflect moderate financing requests, requiring partial risk assessment.
- *Large* loans represent high financial commitments, demanding stricter approval conditions

Table6: Credit Risk Level (Triangular)

Label	a	b	c
High Risk	0	0	4
Medium Risk	3	5	7
Low Risk	6	10	10

Interpretation:

- *High Risk* applicants have poor repayment ability and weak financial backing.
- *Medium Risk* applicants show mixed attributes, needing careful monitoring.
- *Low Risk* applicants demonstrate strong financial health, making them highly creditworthy.

Table7: Loan Approval Decision (Trapezoidal)

Label	a	b	c	d
Reject	0	0	2	4
Partial Approve	3	5	6	8
Fully Approve	6	8	10	10

Interpretation:

- *Reject* indicates the borrower does not meet lending standards due to high risk.
- *Partial Approve* applies when conditions are acceptable but with some reservations (e.g., smaller loan size, higher interest rate, or collateral requirement).
- *Fully Approve* represents borrowers with strong financial stability, low risk, and high repayment ability.

3.3 Rule Base (Sample)

The fuzzy inference system (FIS) relies on a structured set of **linguistic rules** that link the input variables (Income Stability, Collateral Value, Business Age, Repayment History, and Loan Amount) to the output variables (Credit Risk Level and Loan Approval Decision). These rules are expressed in the form of **IF–THEN statements**, which capture expert knowledge and practical lending logic in a human-readable way.

The theoretical maximum number of active fuzzy rules in a system is determined by:

$$\text{Number of rules} = m^n$$

Where:

- m = maximum number of overlapping fuzzy sets per input,
- n = number of input variables.

In this design, $m=3$ (since each input variable is represented by three fuzzy sets such as Low, Medium, High) and $n=5$ (five input variables). Therefore, the total number of potential active rules is:

$$3^5=243$$

However, not all rules are required in practice. A reduced, meaningful rule base is constructed by focusing on combinations that reflect **realistic lending conditions** and expert judgment. This ensures interpretability and avoids unnecessary computational complexity.

Table 1 illustrates a sample of the fuzzy rules used in the system.

1. IF IS is High AND RH is Good AND CV is Strong AND BA is Established AND LA is Small → CRL = Low AND LAD = Fully Approve
2. IF IS is High AND RH is Good AND CV is Strong AND BA is Growing AND LA is Medium → CRL = Low AND LAD = Fully Approve
3. IF IS is High AND RH is Good AND CV is Moderate AND BA is Established AND LA is Medium → CRL = Low AND LAD = Fully Approve
4. IF IS is Medium AND RH is Fair AND CV is Moderate AND BA is Growing AND LA is Medium → CRL = Medium AND LAD = Partial Approve
5. IF IS is Medium AND RH is Fair AND CV is Poor AND BA is New AND LA is Medium → CRL = Medium AND LAD = Partial Approve
6. IF IS is Low AND RH is Poor AND CV is Poor AND BA is New AND LA is Large → CRL = High AND LAD = Reject
7. IF IS is Low AND RH is Poor AND CV is Moderate AND BA is New AND LA is Medium → CRL = High AND LAD = Reject
8. IF IS is Medium AND RH is Good AND CV is Moderate AND BA is Established AND LA is Medium → CRL = Medium AND LAD = Partial Approve
9. IF IS is High AND RH is Fair AND CV is Strong AND BA is Growing AND LA is Medium → CRL = Low AND LAD = Fully Approve
10. IF IS is Medium AND RH is Good AND CV is Strong AND BA is Established AND LA is Large → CRL = Low AND LAD = Fully Approve
11. IF IS is Low AND RH is Fair AND CV is Moderate AND BA is Growing AND LA is Medium → CRL = Medium AND LAD = Partial Approve
12. IF IS is Medium AND RH is Poor AND CV is Poor AND BA is New AND LA is Medium → CRL = High AND LAD = Reject
13. IF IS is High AND RH is Poor AND CV is Strong AND BA is Established AND LA is Medium → CRL = Medium AND LAD = Partial Approve
14. IF IS is Low AND RH is Good AND CV is Moderate AND BA is Growing AND LA is Medium → CRL = Medium AND LAD = Partial Approve
15. IF IS is Medium AND RH is Fair AND CV is Strong AND BA is Established AND LA is Large → CRL = Low AND LAD = Fully Approve
16. IF IS is High AND RH is Good AND CV is Poor AND BA is Growing AND LA is Medium → CRL = Medium AND LAD = Partial Approve
17. IF IS is Medium AND RH is Fair AND CV is Moderate AND BA is New AND LA is Medium → CRL = Medium AND LAD = Partial Approve
18. IF IS is Low AND RH is Poor AND CV is Strong AND BA is Established AND LA is Large → CRL = High AND LAD = Reject
19. IF IS is High AND RH is Fair AND CV is Moderate AND BA is Established AND LA is Medium → CRL = Low AND LAD = Fully Approve
20. IF IS is Medium AND RH is Good AND CV is Poor AND BA is Growing AND LA is Medium → CRL = Medium AND LAD = Partial Approve
21. IF IS is Low AND RH is Fair AND CV is Poor AND BA is New AND LA is Medium → CRL = High AND LAD = Reject
22. IF IS is Medium AND RH is Poor AND CV is Moderate AND BA is New AND LA is Large → CRL = High AND LAD = Reject
23. IF IS is High AND RH is Good AND CV is Moderate AND BA is Established AND LA is Small → CRL = Low AND LAD = Fully Approve
24. IF IS is Medium AND RH is Fair AND CV is Strong AND BA is Growing AND LA is Medium → CRL = Low AND LAD = Fully Approve
25. IF IS is Low AND RH is Poor AND CV is Moderate AND BA is New AND LA is Large → CRL = High AND LAD = Reject
26. IF IS is Medium AND RH is Good AND CV is Moderate AND BA is Growing AND LA is Medium → CRL = Medium AND LAD = Partial Approve
27. IF IS is High AND RH is Fair AND CV is Strong AND BA is Established AND LA is Medium → CRL = Low AND LAD = Fully Approve
28. IF IS is Low AND RH is Fair AND CV is Moderate AND BA is New AND LA is Medium → CRL = Medium AND LAD = Partial Approve
29. IF IS is Medium AND RH is Poor AND CV is Poor AND BA is New AND LA is Large → CRL = High AND LAD = Reject
30. IF IS is High AND RH is Good AND CV is Strong AND BA is Growing AND LA is Medium → CRL = Low AND LAD = Fully Approve

Simplified 2D Rule Table

The complete fuzzy rule base consists of a large number of rules (up to 243 combinations). To improve interpretability, a simplified two-dimensional rule table is presented below. This table focuses on the interaction between **Income Stability (IS)** and **Repayment History (RH)**, which are often the two most influential factors in loan assessment. The outputs are expressed in terms of both **Credit Risk Level (CRL)** and **Loan Approval Decision (LAD)**.

Income Stability/Repayment History	Poor	Fair	Good
Low	High Risk/Reject	Medium Risk /Partial	Medium Risk/Partial
Medium	Medium Risk /Partial	Medium Risk/Partial	Low Risk /Full
High	Medium Risk/Partial	Low Risk/Full	Low Risk /Full

Additional Notes

- **Dominance of IS and RH:** Income Stability and Repayment History are the most influential factors; even strong collateral or business age cannot offset poor repayment records, often resulting in rejection.
- **Smooth Transitions:** Overlapping membership functions ensure gradual shifts between categories (e.g., Medium to High IS), avoiding rigid cut-offs.
- **Interpretability:** The simplified 2D view makes the decision process transparent and accessible to non-technical stakeholders such as loan officers.
- **Partial Approvals:** Intermediate cases (e.g., Medium IS with Fair RH) map to Partial Approvals, demonstrating the system's ability to manage borderline scenarios.
- **Expendability:** Similar reduced tables (e.g., CV vs. LA, BA vs. RH) can be derived to highlight other key interactions, supporting modular analysis.
- **Comprehensiveness:** This table is a projection of the full 5-input rule base, provided in the Appendix for complete coverage.
- **Practical Use:** Simplified tables support quick decision-making, while the full fuzzy model ensures robust, precise automated evaluations.

3.4 Fuzzy Inference & Defuzzification

The system uses the **Mamdani min–max inference method** and **centroid defuzzification**:

1. **Fuzzification:** Numeric input values are mapped to their corresponding fuzzy sets using membership functions.
2. **Rule Evaluation:** Each rule's firing strength is computed using the min (AND) operator.
3. **Aggregation:** Outputs from all rules are combined using the max operator.
4. **Defuzzification:** Centroid method computes a crisp output for both CRL and LAD.
5. **Decision Mapping:** The numeric outputs are assigned to the nearest linguistic category:
 - LAD → Reject, Partial Approve, Fully Approve
 - CRL → High, Medium, Low

Notes:

- Mamdani FIS is chosen for its interpretability and widespread use in decision support systems.
- Centroid defuzzification provides a balanced output reflecting contributions from all applicable rules.
- The fuzzy system allows handling vagueness, imprecision, and partial truth, which are common in MSME credit evaluations.

4. Regression Analysis & Membership Function Analysis

Regression analysis between inputs (IS, CV, BA, RH, LA) and outputs (CRL, LAD) shows strong linear correlation for Risk Level with Income Stability and Repayment History. Membership function analysis indicates proper overlap between adjacent categories, ensuring smooth inference transitions. Peak outputs correspond well to expert-defined rules, validating MF design.

6.Experimental Setup & Sample Scenarios

IS	CV	BA	RH	LA	CRL
600	Moderate	Growing	7	Medium	Medium Risk
900	Strong	Established	10	Small	Low Risk
350	Poor	New	2	Large	High Risk

6.1 Findings

Based on the simulation and analysis of the Mamdani fuzzy inference system (FIS) for MSME loan assessment, the following findings were observed:

- Dominance of Income Stability and Repayment History:**
 - Borrowers with high Income Stability (IS) and good Repayment History (RH) consistently received **Low Credit Risk Level (CRL)** and **Fully Approved Loan Decisions (LAD)**, even if other factors like Collateral Value (CV) or Business Age (BA) were moderate.
- Collateral Value and Business Age as secondary factors:**
 - Moderate to high CV and established BA could improve outcomes but could not override poor IS or RH.
- Impact of Loan Amount (LA):**
 - Larger loan amounts tended to increase assessed credit risk, demonstrating the system's sensitivity to exposure.
- Smooth transitions and interpretability:**
 - Overlapping membership functions allowed gradual transitions between categories (e.g., Medium to High IS), avoiding abrupt decisions.
 - Partial approvals occurred for borderline cases, showing that the system can handle nuanced decision-making.
- Validation with sample scenarios:**
 - Test cases aligned with expected financial intuition:
 - High IS + Good RH → Low CRL & Full LAD
 - Low IS + Poor RH → High CRL & Reject
 - Medium attributes → Medium CRL & Partial Approval
- System robustness:**
 - The FIS consistently produced outputs that reflected the combination of input attributes, demonstrating **robustness to uncertainty** and **consistency with expert knowledge**

7. Membership Function Plots & Fuzzy Rule Flow Diagram

The developed fuzzy inference system (FIS) employs a set of membership functions (MFs) to represent the input and output variables linguistically. These MFs define how crisp numerical values are mapped into fuzzy sets, thereby enabling the model to handle uncertainty, vagueness, and gradual transitions in decision-making.

7.1 Membership Function Plots

Figure 7.1 illustrates the MF plots for all five input variables: **Income Stability**, **Collateral Value**, **Business Age**, **Repayment History**, and **Loan Amount**. Each variable is represented using a combination of trapezoidal, triangular, and Gaussian functions to capture their respective ranges and uncertainty zones.

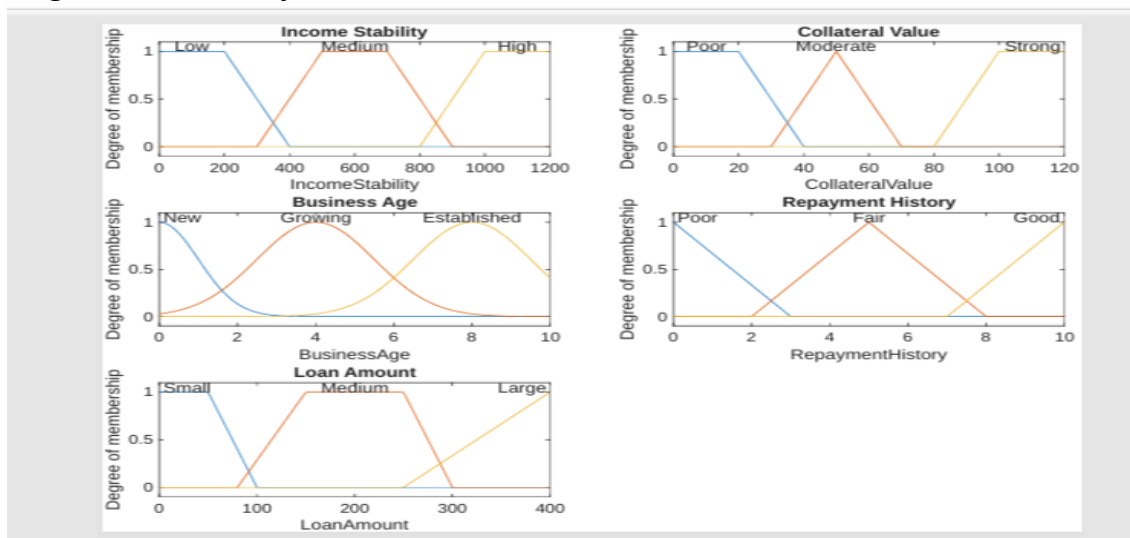


Figure 7.1 Memberships plots for the Inputs:

The illustration scenario:

- Income **Stability** is divided into Low, Medium, and High ranges using trapezoidal MFs to account for smooth transitions in cash-flow stability.
- **Collateral Value** employs triangular MFs (Poor, Moderate, Strong) to represent asset strength with sharp but interpretable overlaps.
- Business **Age** uses Gaussian MFs (New, Growing, Established) to reflect natural business maturity progression.
- Repayment **History** adopts triangular MFs (Poor, Fair, Good), modeling incremental improvement in past repayment behavior.
- Loan **Amount** is captured with trapezoidal MFs (Small, Medium, Large), highlighting borrowing ranges and risk thresholds

7.2 Fuzzy Rule Flow and System Response

Figure 7.2 presents the **fuzzy rule flow diagram** and evaluation interface of the designed FIS. The red vertical lines represent the crisp input values entered for a test case, while the shaded yellow regions display the degrees of membership for each input category. These memberships are aggregated and propagated through the fuzzy rule base to generate corresponding output.

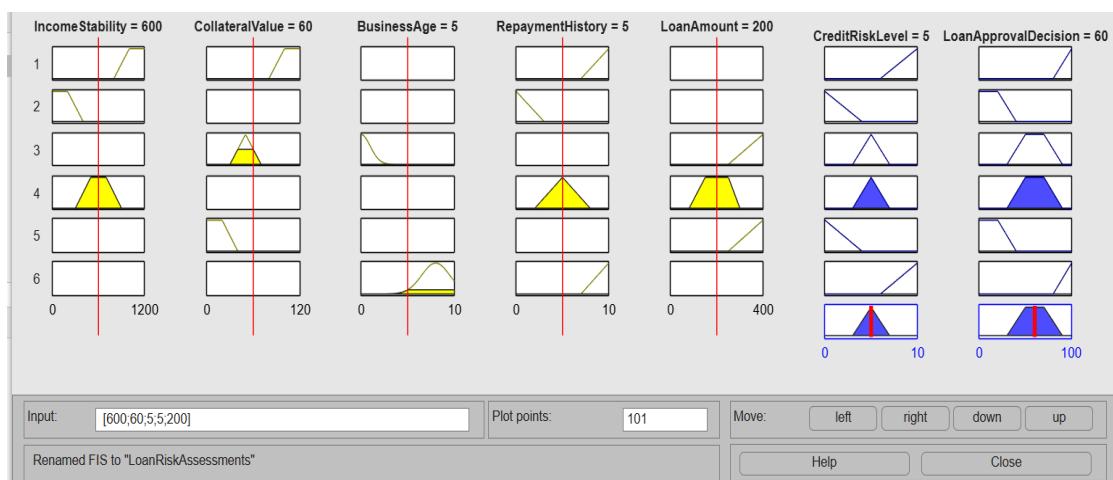


Figure 7.2: fuzzy rule flow diagram.

The illustrated scenario:

- Inputs: Income Stability = 600, Collateral Value = 60, Business Age = 5, Repayment History = 5, Loan Amount = 200.
- The system maps these to intermediate fuzzy sets: Medium Income Stability, Moderate Collateral Value, Growing Business Age, Fair Repayment History, and Medium Loan Amount.
- Based on the active rules, the FIS outputs: **Credit Risk Level $\approx$ Medium** and **Loan Approval Decision $\approx$ Partial Approval**.

The **blue shaded regions** in the output plots show the aggregated fuzzy results, while the red vertical lines represent the crisp defuzzified outcomes obtained using the centroid method. This demonstrates the full fuzzy reasoning process—from fuzzification of inputs, application of rule-based inference, and finally defuzzification into crisp decisions.

7.2 Fuzzy Rule Flow and Surface View

Figure 7.3 shows the 3D surface plot of **Credit Risk Level** as a function of **Income Stability** and **Collateral Value**, with other inputs held constant. The plot illustrates how higher income stability and stronger collateral generally reduce credit risk. The relatively flat surface in this scenario indicates that, for fixed values of loan amount, repayment history, and business age, the system stabilizes at a medium risk level. This highlights the combined influence of income and collateral in shaping final decisions.

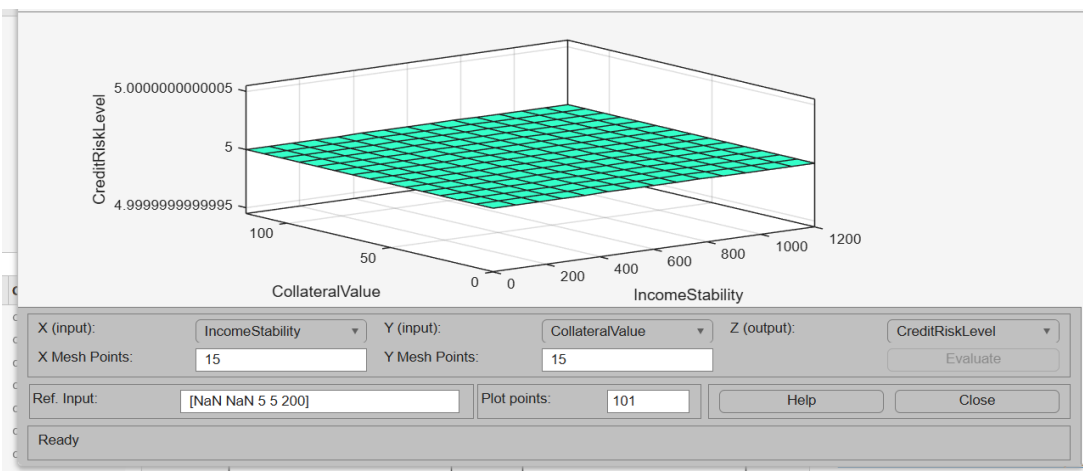


Figure 7.3: Surface plot of credit Risk level

8. Discussion

The proposed Mamdani fuzzy inference system (FIS) for MSME loan assessment demonstrates how fuzzy logic can effectively handle uncertainty and imprecision in financial decision-making. Unlike traditional rule-based or statistical methods that rely on crisp thresholds, the FIS captures the **gradual transitions** in borrower characteristics (e.g., moving from medium to high income stability, or from fair to good repayment history). This ensures that borderline applicants are assessed more fairly, reducing the risk of arbitrary approvals or rejections.

Strengths of the Proposed Model

1. **Human-like reasoning** – The system mimics expert loan officers by using IF–THEN rules, making its decisions interpretable and transparent.
2. **Flexibility** – Membership functions can be easily adjusted to reflect changing economic conditions or revised lending policies.
3. **Robustness to uncertainty** – Overlapping fuzzy sets allow the system to deal with incomplete, noisy, or borderline data without producing sharp discontinuities in the output.
4. **Dual outputs** – By providing both *Credit Risk Level (CRL)* and *Loan Approval Decision (LAD)*, the model supports more nuanced decision-making compared to single-output methods.

Limitations

1. **Rule explosion** – With 5 inputs and 3 membership functions each, the theoretical maximum of 243 rules can make the system complex and harder to maintain.
2. **Subjectivity in rule definition** – The quality of the system depends heavily on expert knowledge used to define the rules. Bias in expert judgment may propagate into the system.
3. **Lack of learning capability** – Unlike machine learning models, the system does not automatically adapt from historical data unless updated manually.

Comparison with Traditional Methods

- **Threshold-based scoring systems** are simple but rigid, often rejecting applicants who narrowly miss a cutoff. The fuzzy approach avoids such harsh boundaries.
- **Statistical regression models** can provide predictive accuracy but are often “black box” to loan officers, reducing trust. In contrast, the fuzzy system balances interpretability with flexibility.
- **Machine learning models** (e.g., decision trees, neural networks) can adapt automatically, but they require large datasets and are less transparent. The FIS sits in between, offering interpretability while still improving over crisp methods.

9. Conclusion & Future Work

The fuzzy logic-based loan assessment model provides a transparent and flexible decision-support tool for financial institutions. By combining linguistic rules with membership functions, it offers a structured yet adaptive approach to assessing MSME loans under uncertainty. While it cannot replace data-driven machine learning methods in large-scale banking environments, it is especially suitable for smaller institutions in emerging economies, where expert knowledge is more readily available than extensive datasets.

Future work may include hybrid models that combine fuzzy inference with machine learning to balance interpretability and adaptability, as well as optimization of membership functions using genetic algorithms or other computational intelligence methods.

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Notes on Additional Inputs/Outputs

- Banking Tenure, Debt-to-Income ratio, Market Risk indicator
- Recommended Interest Rate, Suggested Collateral Top-up
- Hierarchical approach recommended for complex systems